

Multiple choice 2020 - HSC - 2 unit Task 2 (Answers)

Q.1. $y = \sin\left(x \times \frac{\pi}{180}\right)$
 $\frac{dy}{dx} = \frac{\pi}{180} \times \cos \frac{\pi}{180} x$

(C)

Q.2. $\lim_{x \rightarrow 0} \frac{\sin 2x}{3x} = \frac{2}{3}$

(A)

Q.3



$$\int_0^3 f(x) dx = 6$$

$$\int_0^3 f(x) dx = 5 \therefore \int_1^3 f(x) dx = -1$$

(A)

Q.4 $y = 4 \sin\left(\frac{x}{3}\right)$ $T = \frac{2\pi}{\frac{1}{3}} = 6\pi$

(A)

Q.5 $y = \log_e(2x-1) \therefore 2x-1 > 0$
 $x > \frac{1}{2}$

(C)

Q.6 $\int_0^1 (x-1)^2 - (2x-2) dx + \int_1^3 (2x-2) - (x-1)^2 dx$

(D)

Q.7 $f(x) = \ln(\tan x)$

$$f'(x) = \frac{\sec^2 x \rightarrow f'(x)}{\tan x \rightarrow f(x)} = \frac{\frac{1}{\cos^2 x}}{\frac{\sin x}{\cos x}} = \frac{1}{\cos x} \cdot \frac{1}{\sin x} = \sec x \cdot \operatorname{cosec} x$$

(C)

Q.8 $\int_1^d \frac{2}{x} dx = 2 \therefore 2[\ln|x|]_1^d = 2$
 $d > 0$ $\ln d - \ln 1 = 1 \therefore \ln d = 1 \therefore e^1 = d$

(A)

Q.9 $f(x) < 0 \therefore A, C, D$

$f'(x) > 0 \therefore$ increasing $\therefore A, D$

$f''(x) < 0 \therefore$ concave down

$\therefore$ (A)

Q.10 $y = x^2 + bx + c$, $x = \frac{-b}{2a}$ axis passes through vertex which is min. at $x = -2$
 $\therefore -2 = \frac{-b}{2 \times 1} \therefore b = 4$ since $f'(-2) = 0$

now: $y = x^2 + 4x + c$ min. value $(-2, 6)$

$6 = (-2)^2 + 4(-2) + c \therefore c = 10$

(D)

Section II- Extended Response

Attempt Questions 11-16.

Allow about 75 minutes for this section.

Question 11(12 Marks)

a) Differentiate the following

(i) $y = \log_e(2x - 1)^4$

$$y = 4 \log_e(2x - 1)$$

$$\frac{dy}{dx} = 4 \times \frac{2}{2x-1}$$

$$\frac{dy}{dx} = \frac{8}{2x-1}$$

Marking
scale

(2 marks) 2

• correct solus. - (2)

• applies correctly

$\frac{d}{dx} \ln f(x) - (1)$

(ii) $y = \frac{x}{\sin^2 x}$

2

$$u = x \quad u' = 1$$

$$v = \sin^2 x \quad v' = 2 \sin x \cos x$$

$$\therefore \frac{dy}{dx} = \frac{1 \times \sin^2 x - 2x \sin x \cos x}{\sin^4 x}$$

or simplifies further $\frac{dy}{dx} = \frac{\sin x - 2x \cos x}{\sin^3 x}$

(2 marks)

• correct solus - (2)

• applies quotient rule correctly - (1)

(iii) $y = 3^x$

1

$$\frac{dy}{dx} = 3^x \ln 3$$

(1 mark)

• correct solus.

Question 11 continued on the next page

b) Integrate the following

$$\int 3e^{5x+1} - \frac{1}{x^3} dx$$

2

$$= \frac{3}{5} e^{5x+1} - \frac{x^{-2}}{-2} + C$$

(2 marks)

• correct solns.

$$= \frac{3}{5} e^{5x+1} + \frac{1}{2x^2} + C$$

(1 mark)

• Finds the correct primitive of $3e^{5x+1}$

(1 mark)

• Finds the correct primitive of $\frac{1}{x^2}$

c) Evaluate $\int_0^{\frac{\pi}{8}} \sec^2 2x dx$

2

$$= \frac{1}{2} \left[\tan 2x \right]_0^{\pi/8}$$

(2 marks)

• correct solns.

$$= \frac{1}{2} \left[\tan \frac{\pi}{4} - \tan 0 \right]$$

(1 mark)

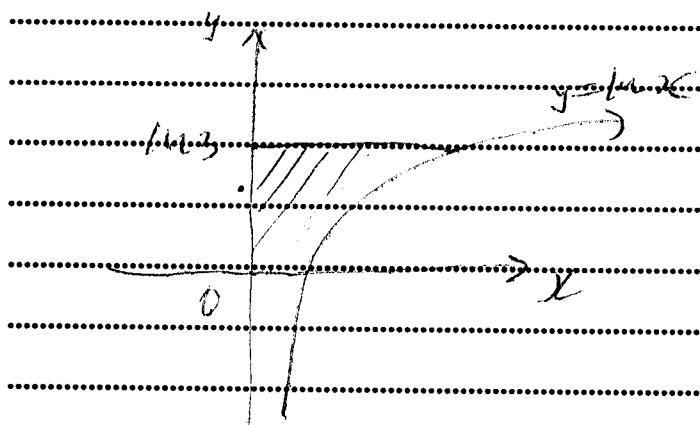
• Find the correct primitive

$$= \frac{1}{2} [1 - 0] = \frac{1}{2}$$

• correctly evaluates definite integral

Question 11 continued on the next page

- d) Find the area bounded by $y = \ln x$, y -axis between $y = 0$ and $y = \ln 3$. 3



(3 marks)

• correct solus.

(2 marks)

• correctly makes x -the subject and establishes the area towards y -axis

$$\text{Area} = \int x \, dy$$

• where $y = \ln x$
 $x = e^y$

$$\therefore \text{Area} = \int_0^{\ln 3} e^y \, dy$$

$$= \left[e^y \right]_0^{\ln 3}$$

$$= e^{\ln 3} - e^0$$

$$= 3 - 1 = 2 \text{ (units}^2\text{)}$$

(1 mark)

• correctly makes x -the subject
 • finds the correct primitive

End of Question 11

Question 12 (7 Marks)

(a)

(i) Differentiate $\cos 3x^3$

→ (2)

$$\frac{d}{dx} \cos 3x^3 = 9x^2 (\sin 3x^3)$$

(2 marks)

• correct solus

$$\therefore \frac{d}{dx} \cos 3x^3 = -9x^2 \sin(3x^3)$$

(1 mark)

• applies chain rule correctly

• correctly differentiates cos-function

(ii) Hence $\int x^2 \sin 3x^3 dx$

2

from (i) $\frac{d}{dx} \cos 3x^3 = -9x^2 \sin 3x^3$

(2 marks)

• correct solus

$$\therefore \cos 3x^3 = \int -9x^2 \sin 3x^3 dx$$

(1 mark)

$$\cos 3x^3 = -9 \int x^2 \sin 3x^3 dx$$

• correctly applies reverse chain rule

$$\therefore \frac{1}{-9} \cos 3x^3 = \int x^2 \sin 3x^3 dx$$

• ignore $+C$

Question 12 continued on the next page

(b) Solve the equation given by $2\ln x = \ln(3 - 2x)$

(3)

$$2 \ln x = \ln(3 - 2x)$$

(3 marks)

correct solns.

$$\ln x^2 = \ln(3 - 2x)$$

(2 marks)

$$x^2 = 3 - 2x$$

• gets both solns.

$$x^2 + 2x - 3 = 0$$

correctly but

$$(x+3)(x-1) = 0$$

doesn't exclude

$$x = -3$$

$$x = -3 \text{ or } x = 1$$

(1 mark)

but $x = -3$ is not a soln. • applies log. rules

since $2\ln(-3)$ is not defined

correctly

$$\therefore \text{soln: } (x=1)$$

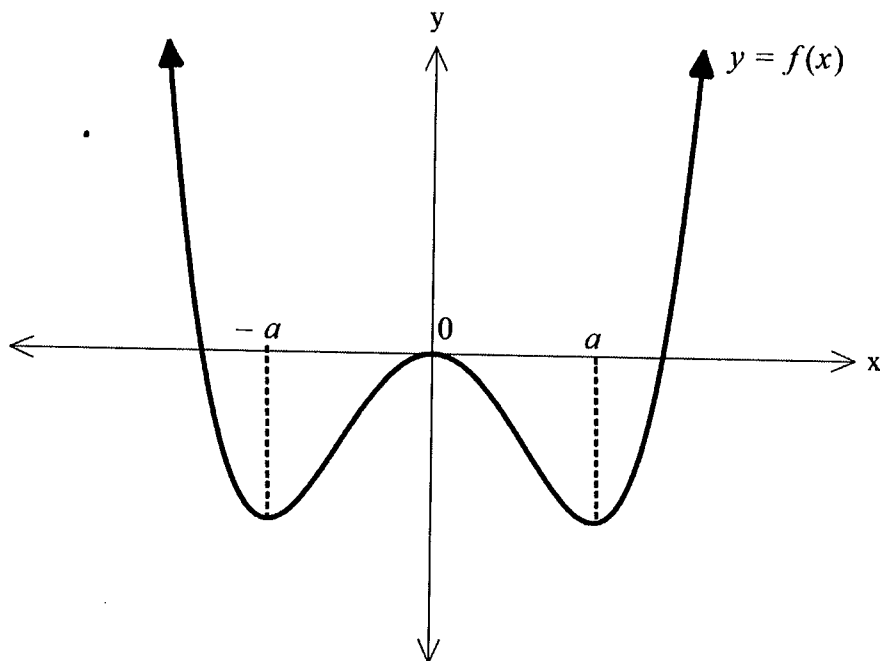
• solves quadratic equation correctly

End of Question 12

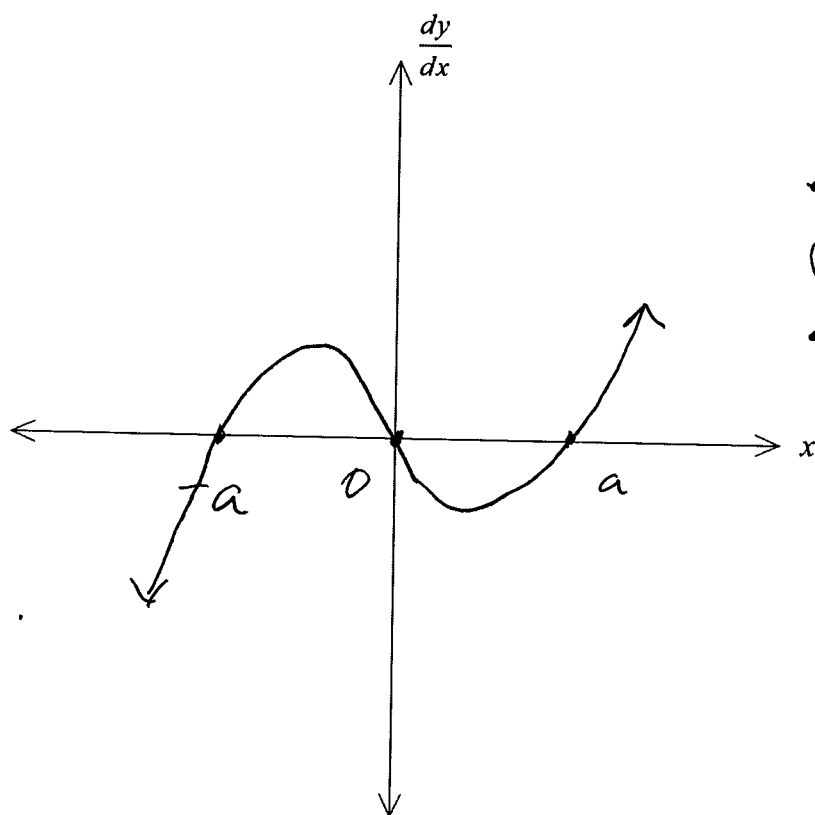
Question 13 (5 Marks)

(a) Given the graph of $y = f(x)$ below

2



Sketch a possible graph of $\frac{dy}{dx}$ on the axes provided.



2 marks

• correct graph

1 mark

• All 3 x-inter. correct

• correct shape

Question 13 continued on the next page

- (b) Water is flowing out of a pool at a rate given by $R = 4t - t^2$ cubic metres/minute. 3
If the volume of water in the pool is initially 10 cubic metres, find the amount of the water in the pool after 4 minutes.

$$R = \frac{dV}{dt}$$

3 marks

• correct solus.

$$\therefore V = \int (4t - t^2) dt$$

2 marks

$$V = -2t^2 + \frac{t^3}{3} + C$$

• finds the correct primitive and evaluates the constant using the initial values

given $t = 0$ $V = 10 \text{ m}^3$

$$\therefore 10 = 2(0) - \frac{0}{3} + C$$

$$\therefore C = 10$$

1 mark

$$\therefore V = -2t^2 + \frac{t^3}{3} + 10$$

• finds the correct primitive
• finds Volumes correctly

Now after $t = 4$ minutes
sub. $t = 4$

$$V = -2(4)^2 + \frac{4^3}{3} + 10$$

$$V = \frac{62}{3} \text{ m}^3 \text{ or } = 20.6 \text{ m}^3$$

$\therefore$ pool is empty

OR

End of Question 13

Question 14 (8 marks)

For the curve given by $f(x) = 2xe^{-x}$

- (i) Show that $f'(x) = 2e^{-x}(1-x)$.

1

$$\begin{aligned}
 f(x) &= 2xe^{-x} \\
 u &= 2x & v &= e^{-x} \\
 u' &= 2 & v' &= -e^{-x} \\
 \therefore f'(x) &= 2e^{-x} + 2x(-e^{-x}) \\
 &= 2e^{-x} - 2xe^{-x} \\
 &= 2e^{-x}(1-x)
 \end{aligned}$$

1 mark
• correct solns.

- (ii) Hence or otherwise find any stationary point(s) and determine their nature.

2

$$\begin{aligned}
 f'(x) &= 0 \\
 2e^{-x}(1-x) &= 0 \\
 2e^{-x} &\neq 0 & 1-x &= 0 \\
 & & x &= 1
 \end{aligned}$$

x	0	1	2
f'(x)	2	0	-0.27

+ / - : (1, 2e⁻¹) Max. turning pt.

2 marks
• correct solns.
1 mark
• solves correctly
• justifies the nature of stat. pt. correctly

- (iii) Determine $\lim_{x \rightarrow \infty} f(x)$.

1

$$\lim_{x \rightarrow \infty} 2xe^{-x} = 0$$

1 mark
• correct solns.

Question 14 continued on the next page

(iv) Find any possible point of inflection.

$$f'(x) = 2e^{-x} - 2xe^{-x}$$

$$f''(x) = -2e^{-x} - (2e^{-x} - 2xe^{-x})$$

2 marks

• correct solns.

1 mark

$$\therefore f''(x) = -2e^{-x} - 2e^{-x} + 2xe^{-x}$$

$$= -4e^{-x} + 2xe^{-x}$$

$$= -2e^{-x}(2 - x)$$

• solves correctly
 $f''(x) = 0$

• shows change in concavity

$$0 = -2e^{-x}(2 - x) \quad \text{at } x = +2$$

$$-2e^{-x} \neq 0 \quad x = +2 \quad \text{possible pt. of inflex.}$$

x	1	+2	3
f''	0.7	0	0.099

Change in concavity

$\therefore (2, 4e^{-2})$ pt. of inflexion

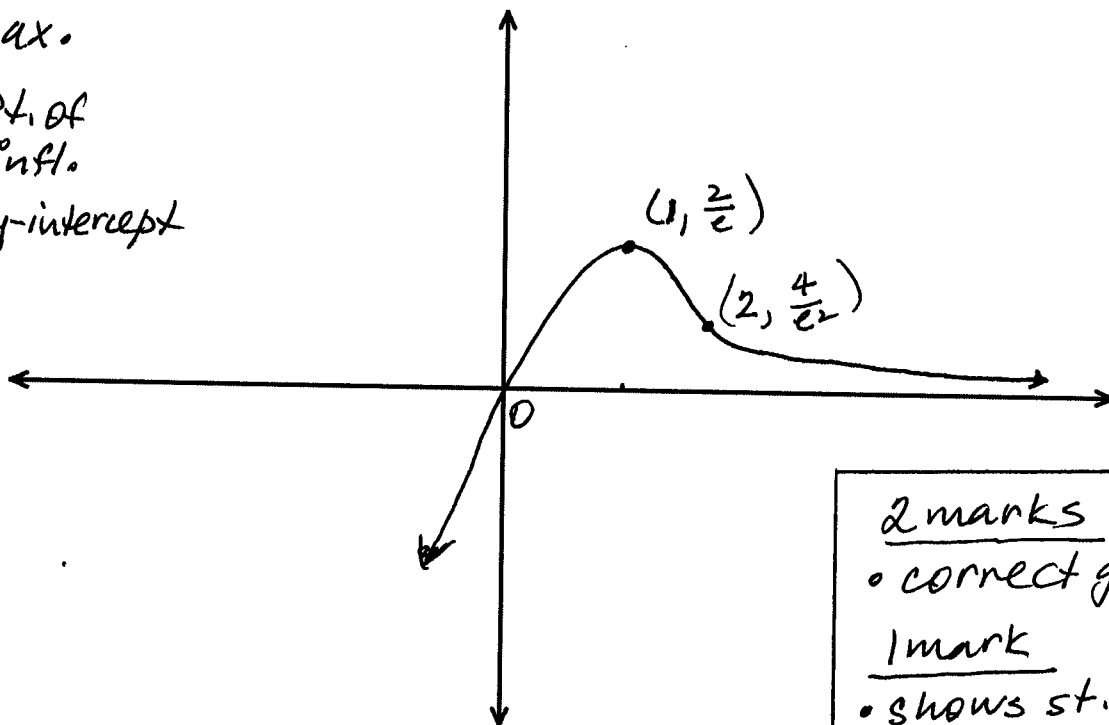
(v) Hence sketch the curve.

2

$(1, \frac{2}{e})$ Max.

$(2, \frac{4}{e^2})$ Pt. of inflex.

$(0, 0)$ - x, y-intercept



2 marks

• correct graph

1 mark

• shows st. point & x-intercept

• correct shape showing asymptote

End of Question 14

Question 15 (10 marks)

The velocity of the particle is given by $v = 2 \sin\left(t - \frac{\pi}{3}\right)$ metres per second.

a) Find:

- (i) the equation of the displacement x of the particle as a function of time if the particle is initially at the origin.

2

$$x = \int v dt$$

$$x = \int 2 \sin\left(t - \frac{\pi}{3}\right) dt$$

$$x = -2 \cos\left(t - \frac{\pi}{3}\right) + C$$

$$\text{at } t=0 \quad x=0$$

$$\therefore 0 = -2 \cos\left(0 - \frac{\pi}{3}\right) + C$$

$$C = 1$$

$$\therefore x = -2 \cos\left(t - \frac{\pi}{3}\right) + 1$$

2 marks

• correct solns.

1 mark

• correctly finds primitive of $2 \sin\left(t - \frac{\pi}{3}\right)$

• substitutes initial conditions
• finds C

- (ii) When does the particle come to the rest first time?

1

$$\text{rest } \therefore v = 0$$

$$0 = 2 \sin\left(t - \frac{\pi}{3}\right)$$

$$t = \frac{\pi}{3} \text{ (seconds)}$$

1 mark

• correct solns.

Question 15 continued on the next page

(iii) the exact initial velocity.

$$t=0 \therefore v = 2 \sin\left(0 - \frac{\pi}{3}\right)$$

$$v = 2 \times \sin\left(-\frac{\pi}{3}\right)$$

$$v = 2 \times -\frac{\sqrt{3}}{2}$$

$$\therefore v = -\sqrt{3} \text{ m/s}$$

1 mark

• correct solns.

(iv) the maximum speed of the particle.

$$\text{speed} = |\text{velocity}|$$

→ velocity graph
is a function of

$$2 \sin\left(t - \frac{\pi}{3}\right)$$

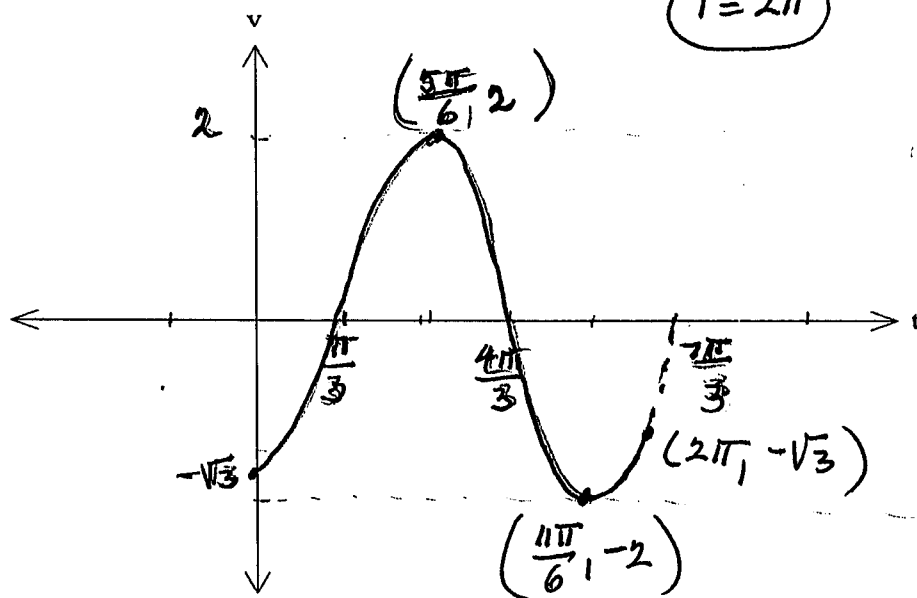
$$\therefore \text{speed}_{\text{max}} = \text{amplitude} = 2 \text{ m/s}$$

1 mark

• correct solns.

(v) Draw the graph of the velocity $v = 2 \sin\left(t - \frac{\pi}{3}\right)$ for $0 \leq t \leq 2\pi$.

Show x, y intercepts and stationary points.



$$T = 2\pi$$

3 marks

• correct graph

2 marks

• correct shape
showing $T = 2\pi$,
and initial $v = -\sqrt{3}$
and zero velocity
at $t = \frac{\pi}{3}$

1 mark

• correct shape
• initial velocity
• $v = 0$ correctly

Question 15 continued on the next page

- (vi) Using your graph above or otherwise, find the total distance travelled by the particle in the first π seconds.

2

$$d = \left| \int_0^{\pi/3} v dt \right| + \int_{\pi/3}^{\pi} v dt$$

2 marks

• correct solus

$$d = \left| \left[-2 \cos \left(t - \frac{\pi}{3} \right) \right]_0^{\pi/3} \right|$$

1 mark

• correctly puts the first section $0 \leq t \leq \frac{\pi}{3}$ into absolute value

$$+ \left[-2 \cos \left(t - \frac{\pi}{3} \right) \right]_{\pi/3}^{\pi}$$

• Finds $\int_0^{\pi} v dt = 2$

$$= \left| -2 (\cos 0 - \cos(\frac{\pi}{3})) \right|$$

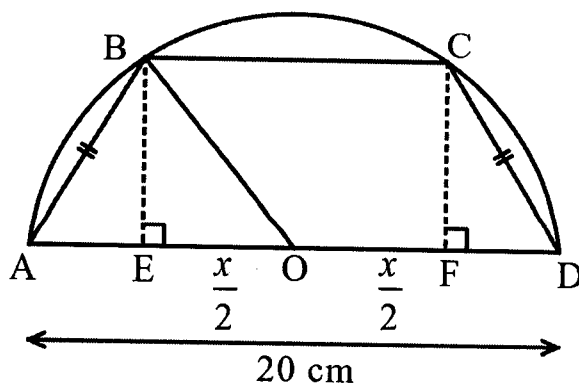
$$+ -2 \left[\cos \frac{2\pi}{3} - \cos 0 \right]$$

$$= 1 + 3 = 4 \text{ metres}$$

End of Question 15

Question 16 (7 Marks)

An isosceles trapezium $ABCD$ is drawn with its vertices on a semicircle centre O and diameter 20 cm. Perpendiculars BE and CF are drawn to meet the diameter AD as shown in the diagram below.



- (i) If $EO = \frac{x}{2}$, show that $BE = \frac{1}{2}\sqrt{400 - x^2}$.

$BO = \text{radius} = 10\text{cm}$

in $\triangle BEO$: $BE^2 = BO^2 - EO^2$

$BE^2 = 10^2 - \left(\frac{x}{2}\right)^2$

$BE^2 = 100 - \frac{x^2}{4} = \frac{400 - x^2}{4}$

$\therefore BE = \frac{1}{2}\sqrt{400 - x^2}$

- (ii) Show that the area of the trapezium $ABCD$ is given by

$A = \frac{1}{4}(x + 20)\sqrt{400 - x^2}$

$BC = x$

$A = \frac{1}{2} \text{height} (BC + AD)$

$\text{height} = BE$ (part(i))

$\therefore A = \frac{1}{2} \times \frac{1}{2} \sqrt{400 - x^2} (x + 20)$

$\therefore A = \frac{1}{4} (x + 20) \sqrt{400 - x^2}$

2 marks
• correct solus.

1 mark

• applies Pythagoras's theorem correctly

2 marks
• correct solus.

1 mark

• applies correctly the area formula of trapezium

• recognises $BC = x$ and height = BE

Question 16 continued on the next page

- (iii) Hence find the length of BC so that the area of the trapezium $ABCD$ is maximum.

3

find $BC = x$

(3 marks)

$$A = \frac{1}{4} (x+20) \sqrt{400-x^2}$$

• correct solus

(2 marks)

$$u = \frac{1}{4} (x+20) \quad v = (400-x^2)^{\frac{1}{2}}$$

• applies product rule correctly and

$$u' = \frac{1}{4}$$

$$v' = \frac{1}{2} x - 2x (400-x^2)^{-\frac{1}{2}} \quad \text{solves for } \frac{dA}{dx} = 0$$

$$v' = -x (400-x^2)^{-\frac{1}{2}}$$

(1 mark)

$$\frac{dA}{dx} = \frac{1}{4} (400-x^2)^{\frac{1}{2}} + \frac{1}{4} (x+20) \cdot -x (400-x^2)^{-\frac{1}{2}}$$

• finds $\frac{dA}{dx}$ correctly

• attempts to solve

$$0 = \frac{1}{4} (400-x^2)^{\frac{1}{2}} - \frac{x(x+20)}{4(400-x^2)^{\frac{1}{2}}}$$

$$\frac{dA}{dx} = 0$$

• find a solution

$$0 = \frac{(400-x^2) - x(x+20)}{4(400-x^2)^{\frac{1}{2}}}$$

$$\text{to } \frac{dA}{dx} = 0$$

justifies the maximum

$$0 = 400 - x^2 - x^2 - 20x$$

$$0 = 2x^2 + 20x - 400$$

$$0 = x^2 + 10x - 200$$

$$0 = (x+20)(x-10)$$

$$x = -20$$

$$x = 10$$

justifies Maximum

but $x > 0$

∴ impossible

End of Exam

x	9	10	11
$\frac{dA}{dx}$	0.81	0	-0.927
	(+)	$x=10$	(-)
	(+)		(-)

∴ at $x = BC = 10\text{cm}$
Area is Maximum



BAULKHAM HILLS HIGH SCHOOL

HSC Assessment Task 2 2020

Mathematics Advanced

General Instructions

- Reading time – 5 minutes
- Working time – 90 minutes
- Write using black pen only
- Board-approved calculators may be used
- Show all necessary working in Questions 11-16
- Marks may be deducted for careless or badly arranged work
- Reference sheet is provided at the end.

Total marks – 59

This paper consists of TWO sections.

Section 1 – Page 2-5 (10 marks)

Questions 1-10

- Attempt questions 1-10
- Allow about 15 minutes for this section

Section II – Pages 6-19 (49 marks)

- Attempt questions 11-19
- Allow about 75 minutes for this section

Section I - 10 marks Answer on the multiple choice page

1. The derivative of $y = \sin x^\circ$ is

(A) $\frac{dy}{dx} = \cos x^\circ$

(B) $\frac{dy}{dx} = -\sin x^\circ$

(C) $\frac{dy}{dx} = \frac{\pi}{180} \cos \frac{\pi}{180} x$

(D) $\frac{dy}{dx} = -\frac{\pi}{180} \cos \frac{\pi}{180} x$

2. $\lim_{x \rightarrow 0} \frac{\sin 2x}{3x} =$

(A) $\frac{2}{3}$

(B) 0

(C) $\frac{3}{2}$

(D) 1

3. The function $y = f(x)$ is continuous for all x .

Given that $\int_0^1 f(x) dx = 6$ and $\int_0^3 f(x) dx = 5$, what is the value of $\int_1^3 f(x) dx$?

(A) -1

(B) 0

(C) 1

(D) 2

4. What is the period of the function $y = 4 \sin\left(\frac{x}{3}\right)$?

(A) 6π

(B) 4

(C) $\frac{2\pi}{3}$

(D) $\frac{1}{4}$

4. The domain for the function $y = \log_e(2x - 1)$ is:

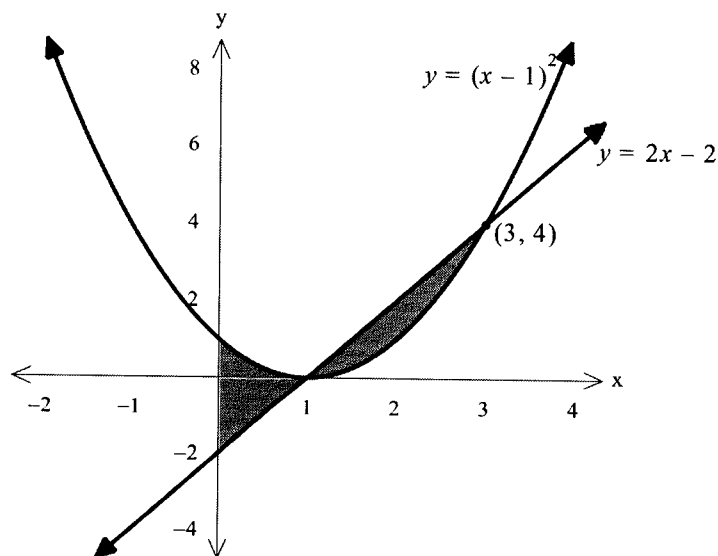
(A) All real $x, x \neq \frac{1}{2}$

(B) $x \geq \frac{1}{2}$

(C) $x > \frac{1}{2}$

(D) $x \leq \frac{1}{2}$

5. The area of the shaded region in the diagram below is given by:



(A) $\int_0^3 2x - 2 - (x - 1)^2 dx$

(B) $\int_1^3 (x - 1)^2 - 2x - 2 dx + \int_0^1 2x - 2 - (x - 1)^2 dx$

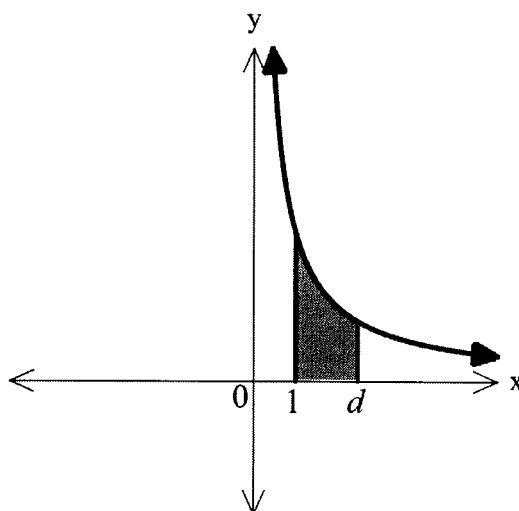
(C) $\int_0^3 2x - 2 - (x - 1)^2 dx$

(D) $\int_0^1 ((x - 1)^2 - 2x + 2) dx + \int_1^3 (2x - 2) - (x - 1)^2 dx$

7. What is $f'(x)$ if $f(x) = \ln(\tan x)$?

- (A) $\frac{1}{\sec^2 x}$
- (B) $\frac{\sec x}{\tan x}$
- (C) $\sec x \cdot \operatorname{cosec} x$
- (D) $\cot x$

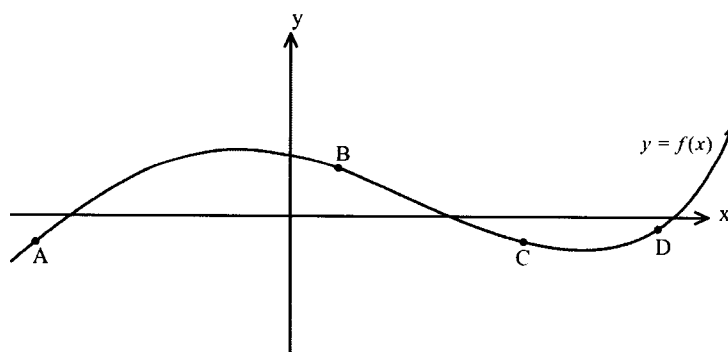
8. The diagram shows the area under the curve $y = \frac{2}{x}$ from $x = 1$ to $x = d$.



What value of d makes the shaded area equal to 2 square units?

- (A) e
- (B) $e + 1$
- (C) $2e$
- (D) e^2

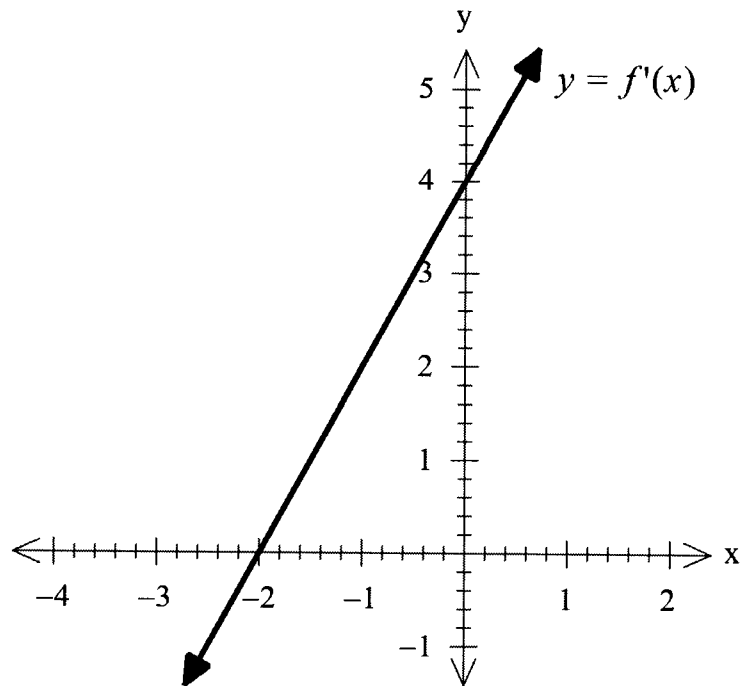
9. The diagram shows the points A, B, C and D on the graph $y = f(x)$



At which point is $f(x) < 0$, $f'(x) > 0$ and $f''(x) < 0$?

- (A) A
- (B) B
- (C) C
- (D) D

10. The graph of $y = f'(x)$ is given below.



The curve $y = f(x)$ has a minimum value of 6. What is the equation of the curve ?

- (A) $y = x^2 - 4x + 2$
- (B) $y = x^2 - 4x + 10$
- (C) $y = x^2 + 4x + 2$
- (D) $y = x^2 + 4x + 10$

END OF SECTION I

Section II- Extended Response

Attempt Questions 11-16.

Allow about 75 minutes for this section.

Question 11(12 Marks)

(a) Differentiate the following

(i) $y = \log_e(2x - 1)^4$ 2

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(ii) $y = \frac{x}{\sin^2 x}$ 2

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(iii) $y = 3^x$ 1

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Question 11 continued on the next page

b) Integrate the following

$$\int 3e^{5x+1} - \frac{1}{x^3} dx$$

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c) Evaluate $\int_0^{\frac{\pi}{8}} \sec^2 2x dx$

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Question 11 continued on the next page

Question 12 (7 Marks)

(a)

(i) Differentiate $\cos 3x^3$

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(ii) Hence find $\int x^2 \sin 3x^3 dx$

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Question 12 continued on the next page

(b) Solve the equation given by $2\ln x = \ln(3 - 2x)$

3

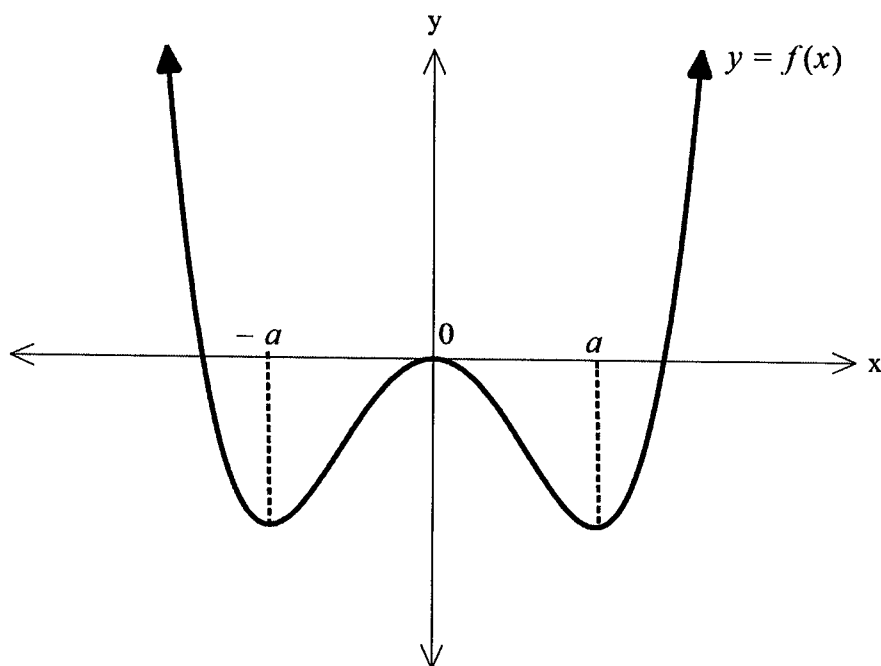
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End of Question 12

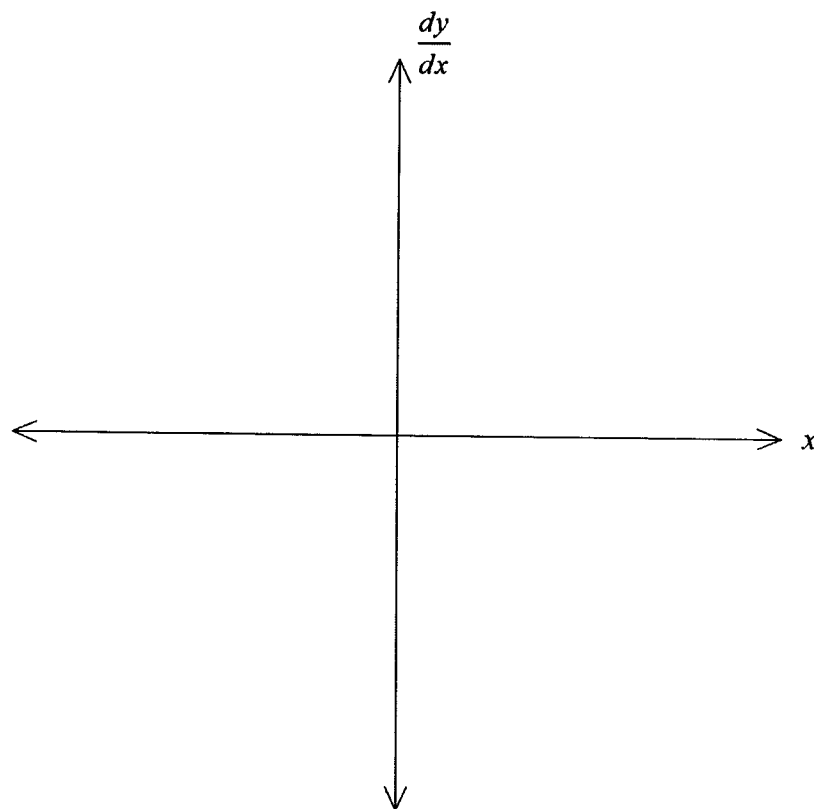
Question 13 (5 Marks)

(a) Given the graph of $y = f(x)$ below

2



Sketch a possible graph of $\frac{dy}{dx}$ on the axes provided.



Question 13 continued on the next page

(b) Water is flowing out of a pool at a rate given by $R = 4t - t^2$ cubic metres/minute 3

If the volume of water in the pool is initially 10 cubic metres, find the amount of the water in the pool after 4 minutes.

[illegible]

End of Question 13

Question 14 (8 marks)

For the curve given by $f(x) = 2xe^{-x}$

- (i) Show that $f'(x) = 2e^{-x}(1 - x)$. 1

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- (ii) Hence or otherwise find any stationary point(s) and determine their nature. 2

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- (iii) Determine $\lim_{x \rightarrow \infty} f(x)$. 1

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Question 14 continued on the next page

(iv) Find any possible point of inflection.

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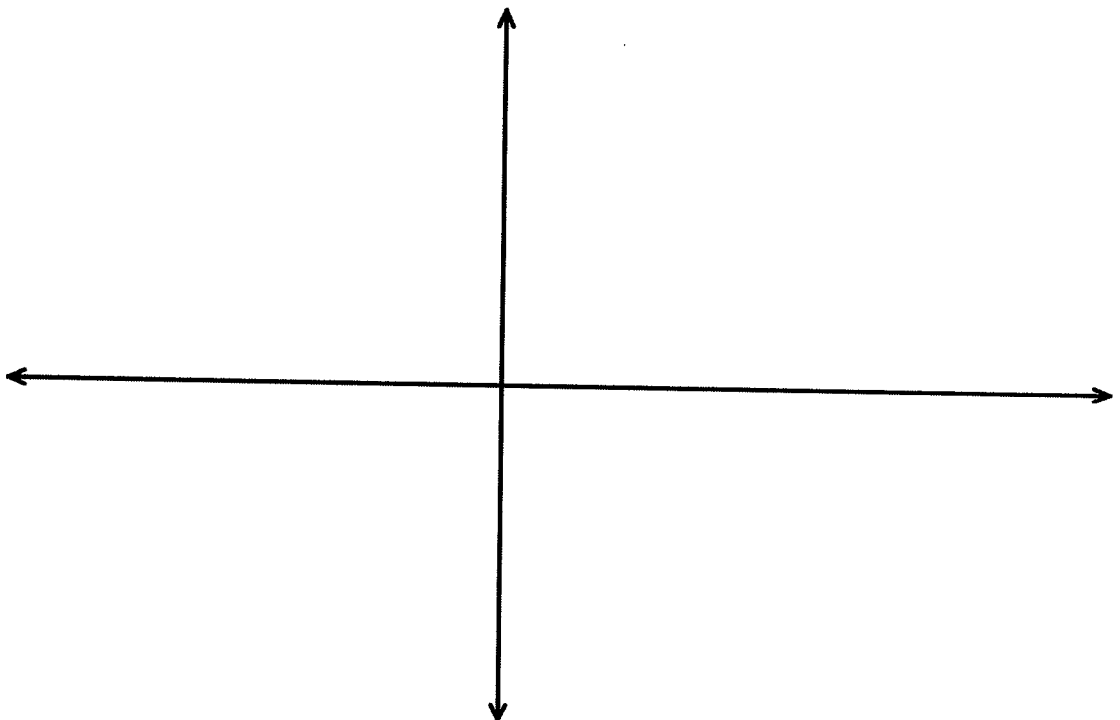
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(v) Hence sketch the curve.

2



End of Question 14

Question 15 (10 marks)

The velocity of a particle is given by $v = 2 \sin\left(t - \frac{\pi}{3}\right)$ metres per second.

Find:

- (i) the equation of the displacement x of the particle as a function of time if the particle is initially at the origin. **2**

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- (ii) when the particle comes to rest for first time. **1**

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Question 15 continued on the next page

(iii) the exact initial velocity.

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(iv) the maximum speed of the particle.

1

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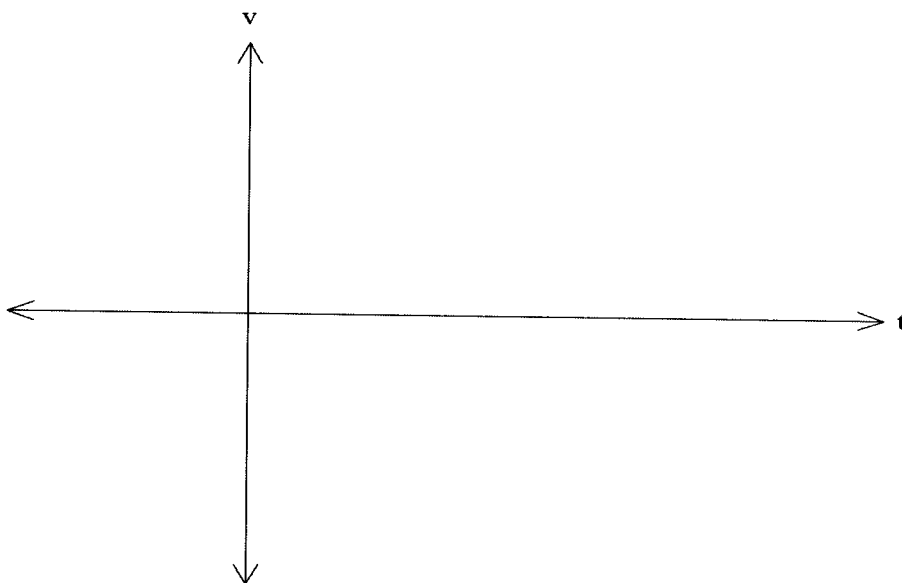
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(v) Draw the graph of the velocity $v = 2\sin\left(t - \frac{\pi}{3}\right)$ for $0 \leq t \leq 2\pi$.

3

Show all intercepts and stationary points.



Question 15 continued on the next page

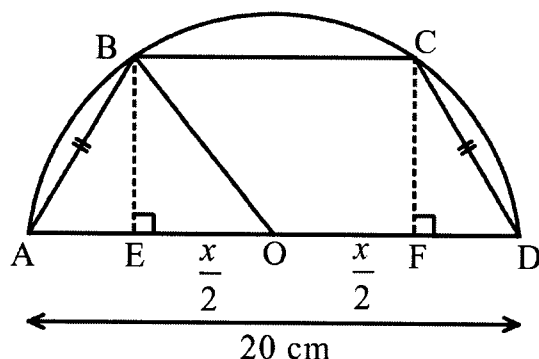
- (vi) Using your graph above or otherwise, find the total distance travelled by the particle in the first π seconds.

[illegible]

End of Question 15

Question 16 (7 Marks)

An isosceles trapezium $ABCD$ is drawn with its vertices on a semicircle centre O and diameter 20 cm. Perpendiculars BE and CF are drawn to meet the diameter AD as shown in the diagram below.



- (i) If $EO = OF = \frac{x}{2}$, show that $BE = \frac{1}{2}\sqrt{400 - x^2}$. 2

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- (ii) Show that the area of the trapezium $ABCD$ is given by

$$A = \frac{1}{4}(x + 20)\sqrt{400 - x^2} . \quad \quad \quad 2$$

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Question 16 continued on the next page

- (iii) Hence find the length of BC so that the area of the trapezium $ABCD$ is maximum.

3

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

End of Exam



BAULKHAM HILLS HIGH SCHOOL

YEAR 12 – MATHEMATICS

NESA# _____

Mathematics

Section I – Multiple Choice

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct

- Start Here** →
- | | | | | | | | |
|----------------------------|-------------------------|-------------------------|-------------------------|-----------------------------|-------------------------|-------------------------|-------------------------|
| 1. A ☐ | B ☐ | C ☐ | D ☐ | 6. A ☐ | B ☐ | C ☐ | D ☐ |
| 2. A ☐ | B ☐ | C ☐ | D ☐ | 7. A ☐ | B ☐ | C ☐ | D ☐ |
| 3. A ☐ | B ☐ | C ☐ | D ☐ | 8. A ☐ | B ☐ | C ☐ | D ☐ |
| 4. A ☐ | B ☐ | C ☐ | D ☐ | 9. A ☐ | B ☐ | C ☐ | D ☐ |
| 5. A ☐ | B ☐ | C ☐ | D ☐ | 10. A ☐ | B ☐ | C ☐ | D ☐ |